

Review of Probability and Statistics

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Random Variables and Probability Distributions

Probability Theory

The theory of probability is the basic language of uncertainty and also forms the basis for statistical inference.

Outcomes

The mutually exclusive potential results of a random process are called outcomes. For example, if you shoot a penalty kick in football (soccer) possible outcomes are: i) you score a goal, ii) you do not score a goal.

Sample Space

The set of all possible outcomes is called the sample space.

Event

An event is a subset of the sample space, that is, an event is a set containing one or more outcomes.

Random Variables and Probability Distributions

Probability of an Outcome

The probability of an outcome is the proportion of the time that the outcome occurs in the long run. In other words, if an experiment has been repeated over a huge number of trials, we can approximate the probability of an outcome by using the relative frequency.

Probability of an Event

The probability of an event is the sum of the probabilities of the outcomes in the event.

Summary from last class

- Basic structure: the notion of a probability space:

$$(S, \Sigma, \mathbb{P})$$

where

- S is the set of outcomes $\rightarrow$ mutually exclusive enumeration of how uncertainty is resolved.
- Σ is the set of events: a collection of subsets of S , it describes formally which collections of outcomes we are allowed to “talk about”.
- $\mathbb{P}$, a probability function, that assigns a number, to each event in Σ , interpreted as the likelihood of it happening.
- A random variable: a function of from S to the real numbers.

$$X : S \rightarrow \mathbb{R}$$

- Notice, the notation, the random variable (function) in capital letters.
- A particular value of the random variable, lower case letter

$$X(s) = x, \text{ where } s \in S$$

Random Variables and Probability Distributions

Random Variable

A random variable is a numerical summary of a random outcome. Random variables can be discrete or continuous.

Probability Distribution

The probability distribution of a discrete random variable is the list of all possible values of the variable and the probability that each value will occur. The probability of an event can be computed from the probability distribution.

Cumulative Probability Distribution

The cumulative probability distribution (also cumulative distribution function or c.d.f.) is the probability that the random variable is less than or equal to a particular value.

Random Variables and Probability Distributions

Example

- Experiment: Flip a coin four times.
- Define the random variable X to be the number of tails.
- Write down the probability distribution.
- Write down the cumulative probability distribution.
- Compute the probability of the event X being even.

Four coin tosses

$$S = \begin{pmatrix} H & H & H & H \\ H & H & H & T \\ H & H & T & H \\ H & H & T & T \\ H & T & H & H \\ H & T & H & T \\ H & T & T & H \\ H & T & T & T \\ T & H & H & H \\ T & H & H & T \\ T & H & T & H \\ T & H & T & T \\ T & T & H & H \\ T & T & H & T \\ T & T & T & H \\ T & T & T & T \end{pmatrix}$$

- Σ , the set of events is too large (all subsets of S): its size is 2^{16} .
- $\mathbb{P}$: notice that each outcome in S has the same probability if the coin is fair, $1/16$. With this we can construct the probability of any event we want to discuss $\mathbb{P}(\text{"at least three heads"}) = \mathbb{P}\{HHHH, HHHT, HHTH, HTHH, THHH\} = 5/16$

X : number of tails

- X can take on five values: $\{0, 1, 2, 3, 4\}$, it counts the number of tails.
- $\mathbb{P}$ will induce a probability distribution on those five values.
- $\Pr(X = 0) = \mathbb{P}(HHHH) = 1/16$.
- $\Pr(X = 1) = \mathbb{P}(HHHT, HHTH, HTHH, THHH) = 4/16 = 1/4$
- and so on...

Random Variables and Probability Distributions

The Bernoulli Distribution

A binary random variable is called a Bernoulli random variable and its probability distribution is called the Bernoulli distribution. For example, let G denote the gender of the next person you meet, where $G = 0$ indicates the person is male and $G = 1$ indicates that she is female.

$$G = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases} \quad (1)$$

The Binomial distribution

The Binomial Distribution

The binomial distribution is the discrete probability distribution of the number of successes in a sequence of n independent yes/no experiments, i.e. n Bernoulli trials, where the probability of success is p

$$\Pr(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \quad (2)$$

Random Variables and Probability Distributions

Probability Density Function of a Continuous Random Variable

The area under the probability density function (also called p.d.f., or density) between any two points is the probability that the random variable falls between those two points.

CDF for a Continuous Random Variable

The cumulative probability distribution for a continuous random variable is defined as it is for a discrete random variable. The cumulative probability distribution is the probability that the random variable is less than or equal to a particular value.

Random Variables and Probability Distributions

The Uniform Distribution

$$f(y) = \begin{cases} \frac{1}{b-a} & \text{for } y \in [a, b] \\ 0 & \text{otherwise} \end{cases} \quad (3)$$

where a and b are known constants.

Expected Values, Mean, and Variance

Expected Value of a Discrete Random Variable

The expected value of a random variable Y , denoted by $E[Y]$ or μ_Y is the long run average value of the random variable over many repeated trials or occurrences. The expected value is computed as a weighted average of the possible outcomes of that random variable where the weights are the outcomes' probabilities.

Some Notation

Suppose the random variable Y takes on K possible values $y_1, \dots, y_K$ with probabilities $p_1, \dots, p_K$ respectively. The expected value of Y , denoted by $E[Y]$ is:

$$E[y] = \mu_Y = p_1 y_1 + p_2 y_2 + \dots + p_K y_K = \sum_{i=1}^K p_i y_i \quad (4)$$

Expected Values, Mean, and Variance

Expected Value of a Continuous Random Variable

A continuum random variable can take on an uncountable infinite many possible values. The "summation notation" we introduced before will not work. We need to replace the summation sign by the integral sign.

$$E[y] = \mu_Y = \int y f_Y(y) dy. \quad (5)$$

where the range of integration is the set of values of Y for which its density $f_Y(y)$ is nonzero.

Expected Values, Mean, and Variance

Standard Deviation and Variance

The standard deviation and variance are measures of dispersion of a probability distribution. The variance of the random variable Y , denoted by $\text{Var}[Y]$ or σ_Y^2 , is the expected value of the square of the deviation of Y from its mean. The standard deviation is simply the square root of the variance and is denoted by σ_Y . Hence

$$\text{Var}[Y] = \sigma_Y^2 = E \left[(Y - \mu_Y)^2 \right] = \sum_{i=1}^K p_i (y_i - \mu_Y)^2, \quad (6)$$

$$\text{Stdev}[Y] = \sigma_Y = [\text{Var}[Y]]^{\frac{1}{2}}. \quad (7)$$

Expected Values, Mean, and Variance

Mean and Variance of a Linear Function of a Random Variable

Let Y be given by $Y = a + bX$. Then, the mean and variance of Y are given by

$$E[Y] = \mu_Y = a + b\mu_X, \quad \text{Var}[Y] = \sigma_Y^2 = b^2 \sigma_X^2. \quad (8)$$

Using the results in Equation (8) we can write $\text{Var}[Y]$ as

$$\text{Var}[Y] = \sigma_Y^2 = E[Y^2] - [E[Y]]^2. \quad (9)$$

Expected Values, Mean, and Variance

Other Measures of the Shape of a Distribution

The mean and the standard deviation measure the center of the distribution and its spread. The skewness of the distribution measures the lack of symmetry of a of a distribution. The kurtosis measures how "thick" or "heavy" the tails of the distribution are.

Expected Values, Mean, and Variance

Skewness

$$\text{Skewness} = \frac{E \left[(Y - \mu_Y)^3 \right]}{\sigma_Y^3}. \quad (10)$$

- For a symmetric distribution skewness is zero.
- If the distribution has long right tale, the skewness is positive.
- If the distribution has long left tale, the skewness is negative.

Expected Values, Mean, and Variance

Kurtosis

$$\text{Kurtosis} = \frac{E \left[(Y - \mu_Y)^4 \right]}{\sigma_Y^4}. \quad (11)$$

- The greater the kurtosis the more likely outliers are. For a normally distributed random variable the kurtosis is 3.
- A random variable with kurtosis larger than 3 (leptokurtic) has more mass in its tail than a normal random variable.

Expected Values, Mean, and Variance

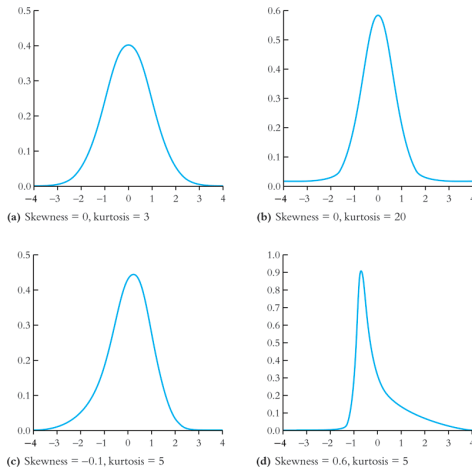


Figure : Skewness and Kurtosis

Expected Values, Mean, and Variance

Moments of a Distribution

- The mean of Y , $E[Y]$, is the first moment of Y ; the expected value of Y^2 , $E[Y^2]$, is the second moment of Y .
- In general the expected value of Y^r is the r^{th} moment of the random variable Y .
- The variance is a function of the first and second moments, the skewness is a function of the first, second, and third moments, and the kurtosis is a function of the firsts through fourth moments of Y .

Two Random Variables

Joint Probability Distribution

The joint probability distribution of two discrete random variables X and Y is the probability that the random variables simultaneously take on certain values x and y . The joint probability distribution can be denoted by

$$f(x, y) = \Pr(X = x, Y = y). \quad (12)$$

Two Random Variables

Marginal Probability Distribution

The marginal probability distribution of a random variable Y is the probability distribution of Y . It can be computed from the joint distribution of X and Y by adding up the probabilities of all the outcomes for which Y takes on a specific value, repeating the procedure for all possible values of Y).

$$f_Y(x, y) = Pr(Y = y) = \sum_{i=1}^I Pr(X = x_i, Y = y). \quad (13)$$

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table a: Joint Probability Distribution

- Calculate $P(X=5, Y=40)$
- Calculate $P(Y=30)$
- Compute the marginal probability distribution of X and Y.
- Compute the mean, variance, and standard deviation of Y

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table a: Joint Probability Distribution

- Calculate $P(X=5, Y=40) = .02$
- Calculate $P(Y=30)$
- Compute the marginal probability distribution of X and Y.
- Compute the mean, variance, and standard deviation of Y

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table a: Joint Probability Distribution

- Calculate $P(X=5, Y=40) = .02$
- Calculate $P(Y=30) = 0.1 + 0.05 + 0.15 = 0.3$
- Compute the marginal probability distribution of X and Y.
- Compute the mean, variance, and standard deviation of Y

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table a: Joint Probability Distribution

- Calculate $P(X=5, Y=40) = .02$
- Calculate $P(Y=30) = 0.1 + 0.05 + 0.15 = 0.3$
- Compute the marginal probability distribution of X and Y.
 $f_X(x) = [0.21 \quad 0.4 \quad 0.39]$, $f_Y(y) = [0.2100 \quad 0.2300 \quad 0.3000 \quad 0.1500 \quad 0.1100]$
- Compute the mean, variance, and standard deviation of Y

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table a: Joint Probability Distribution

- Calculate $P(X=5, Y=40) = .02$
- Calculate $P(Y=30) = 0.1 + 0.05 + 0.15 = 0.3$
- Compute the marginal probability distribution of X and Y.
 $f_X(x) = [0.21 \quad 0.4 \quad 0.39]$, $f_Y(y) = [0.2100 \quad 0.2300 \quad 0.3000 \quad 0.1500 \quad 0.1100]$
- Compute the mean, variance, and standard deviation of Y
 $\mu_X = 30.15$, $\sigma_Y^2 = 218.2075$, $std(Y) = 14.7718$

Two Random Variables

Conditional Distribution

The conditional distribution of a random variable Y is the distribution of Y conditional on X taking a specific value x . The conditional distribution can be denoted by:

$$Pr(Y = y|X = x) = \frac{Pr(X = x, Y = y)}{Pr(X = x)}. \quad (14)$$

Two Random Variables

Conditional Expectation

The conditional expectation of Y given X is the mean of the conditional distribution of Y given X . In other words, it is the expected value of Y computed using the conditional distribution of Y given X . If Y takes on K values $y_1, \dots, y_K$, the conditional mean of Y given $X = x$ is

$$E[Y|X = x] = \sum_{i=1}^K Pr(Y = y_i|X = x) y_i. \quad (15)$$

Two Random Variables

Law of Iterated Expectation

The expected value of Y is the weighted average of the conditional expectation of Y given X , weighted by the probability distribution of X . Mathematically, if X takes on the L values $x_1, \dots, x_L$ then

$$E[Y] = \sum_{i=1}^L Pr(X = x_i) E[Y|X = x_i]. \quad (16)$$

In other words, the expected value of Y is equal to the expectation of the conditional expectation of Y given X

$$E[Y] = E[E[Y|X]]. \quad (17)$$

Equation (17) is called the *Law of Iterated Expectations (LIE)*

Two Random Variables

- The inner expectation in (LIE) is computed using the conditional distribution of Y given X , and the outer expectation is computed using the marginal distribution of X .
- The (LIE) implies that if the conditional mean of Y given X is zero, then the mean of Y is zero.
- The (LIE) also applies to expectations that are conditioned on multiple random variables.

Two Random Variables

Conditional Variance

The conditional variance of Y given X is the variance of the conditional distribution of Y given X . Mathematically

$$\begin{aligned} V[Y|X = x] &= E[(Y - E[Y|x])^2 | X = x], \\ &= \sum_{i=1}^K \Pr(Y = y_i | X = x) [y_i - E[Y|X = x]]^2. \end{aligned} \quad (18)$$

As before

$$V[Y|X = x] = E[Y^2|X = x] - (E[Y|X = x])^2. \quad (19)$$

Two Random Variables

Independence

Two random variables X and Y are independently distributed (or independent) if knowing the value of one of the variables provides no information about the other. Specifically, X and Y are independent if the conditional distribution of Y given X equals the marginal distribution of Y . This is

$$Pr(Y = y|X = x) = Pr(Y = y). \quad (20)$$

Using the definition of independence into the formula for the conditional distribution, we find an expression for the joint distribution of two independent variables

$$Pr(Y = y, X = x) = Pr(Y = y) Pr(X = x). \quad (21)$$

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table b: Joint Probability Distribution

- Calculate the probability distribution of Y conditional on $X = 8$, $P(Y = y|X = 8)$
- Compute the mean of Y conditional on $X = 8$, $E[Y|X = 8]$
- Compute the variance and standard deviation of Y conditional on $X = 8$, $V[Y|X = 8]$

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table b: Joint Probability Distribution

- Calculate the probability distribution of Y conditional on $X = 8$, $P(Y = y|X = 8)$

$$f_{Y|X=8}(y) = [0.0513 \quad 0.0769 \quad 0.3846 \quad 0.2564 \quad 0.2308]$$

- Compute the mean of Y conditional on $X = 8$, $E[Y|X = 8]$
- Compute the variance and standard deviation of Y conditional on $X = 8$, $V[Y|X = 8]$

Two Random Variables

Example

		Value of Y				
		14	22	30	40	65
Value of X	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table b: Joint Probability Distribution

- Calculate the probability distribution of Y conditional on $X = 8$, $P(Y = y|X = 8)$

$$f_{Y|X=8}(y) = [0.0513 \quad 0.0769 \quad 0.3846 \quad 0.2564 \quad 0.2308]$$

- Compute the mean of Y conditional on $X = 8$, $E[Y|X = 8] = 39.2051$
- Compute the variance and standard deviation of Y conditional on $X = 8$, $V[Y|X = 8]$

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table b: Joint Probability Distribution

- Calculate the probability distribution of Y conditional on $X = 8$, $P(Y = y|X = 8)$

$$f_{Y|X=8}(y) = [0.0513 \quad 0.0769 \quad 0.3846 \quad 0.2564 \quad 0.2308]$$

- Compute the mean of Y conditional on $X = 8$, $E[Y|X = 8] = 39.2051$
- Compute the variance and standard deviation of Y conditional on $X = 8$, $V[Y|X = 8] = 323.6456$, $std_{Y|X=8} = 17.92$.

Two Random Variables

Covariance and Correlation

The covariance between two variables is the extent to which the two variables move together. If X can take on L values and Y can take on K values, then the covariance is

$$\begin{aligned} \text{Cov}(X, Y) &= E[(X - \mu_X)(Y - \mu_Y)], \\ &= \sum_{i=1}^K \sum_{j=1}^L \text{Pr}(Y = y_i, X = x_j) (y_i - \mu_Y)(x_j - \mu_X), \\ &= \sigma_{XY}. \end{aligned} \tag{22}$$

As in the case for the variance, we can write

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]. \tag{23}$$

Two Random Variables

Covariance and Correlation

$$\text{Cov}(X, Y) = \sigma_{XY} = E[(X - \mu_X)(Y - \mu_Y)].$$

- Suppose that when the value of X is greater than its mean (so $(X - \mu_X) > 0$) the value of Y tends to be greater than its mean (so $(Y - \mu_Y) > 0$), and when X is less than its mean (so $(X - \mu_X) < 0$) the value of Y tends to be less than its mean (so $(Y - \mu_Y) < 0$). Then, the product $(X - \mu_X)(Y - \mu_Y)$ tends to be positive, and the covariance is positive.
- In contrast if X and Y tend to move in opposite directions, then the covariance is negative.
- Finally, if X and Y are independent, then the covariance is zero.

Two Random Variables

Covariance and Correlation

The correlation is an alternative measure of dependence between X and Y . The correlation coefficient, unlike the covariance, is unitless. The correlation coefficient between X and Y is

$$\text{Corr}(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}. \quad (24)$$

- The random variables X and Y are said to be uncorrelated if $\text{Corr}(X, Y) = 0$
- The correlation coefficient is always less than one in absolute value: $-1 \leq \text{Corr}(X, Y) \leq 1$.

Two Random Variables

Correlation and Conditional Mean

If the conditional mean of Y does not depend on X , then Y and X are uncorrelated

$$\text{If } E[Y|X] = \mu_Y \longrightarrow \text{Cov}(X, Y) = 0 \longrightarrow \text{Corr}(X, Y) = 0 \quad (25)$$

It is not necessarily true, however, that if Y and X are uncorrelated, then the conditional mean of Y given X does not depend on X .

Two Random Variables

Means, Variances, and Covariances of Sums of Random Variables

Let X , Y , and V be random variables and a , b , and c constants. Then

$$E[a + bX + cY] = a + b\mu_X + c\mu_Y, \quad (26)$$

$$V[a + bY] = b^2\sigma_Y^2, \quad (27)$$

$$V[aX \pm bY] = a^2\sigma_X^2 + b^2\sigma_Y^2 \pm 2ab\sigma_{XY}, \quad (28)$$

$$E[Y^2] = \sigma_Y^2 + \mu_Y^2, \quad (29)$$

$$\text{Cov}[a + bX + cV, Y] = b\sigma_{XY} + c\sigma_{VY}, \quad (30)$$

$$E[XY] = \sigma_{XY} + \mu_X\mu_Y, \quad (31)$$

$$|\text{Corr}(X, Y)| \leq 1 \text{ and } |\sigma_{XY}| \leq \sqrt{\sigma_X^2\sigma_Y^2}. \quad (32)$$

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table c: Joint Probability Distribution

- Calculate the covariance and correlation between X and Y ,
 $Cov(X, Y)$, $Corr(X, Y)$.

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table c: Joint Probability Distribution

- Calculate the covariance and correlation between X and Y ,
 $Cov(X, Y) = 11.0005$, $Corr(X, Y)$.

Two Random Variables

Example

		Value of Y				
Value of X		14	22	30	40	65
	1	0.02	0.05	0.1	0.03	0.01
	5	0.17	0.15	0.05	0.02	0.01
	8	0.02	0.03	0.15	0.10	0.09

Table c: Joint Probability Distribution

- Calculate the covariance and correlation between X and Y ,
 $Cov(X, Y) = 11.0005$, $Corr(X, Y) = 0.2864$.

Some Important Distributions

The Normal Distribution

- The normal distribution is a continuous distribution that has the familiar bell-shaped curve that is symmetric around the mean and has 95% of its probability between $\mu - 1.96\sigma$ and $\mu + 1.96\sigma$
- The normal distribution with mean μ and variance σ^2 is denoted by $N(\mu, \sigma^2)$. The standard normal distribution is the normal distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$ and is denoted by $N(0, 1)$.

Some Important Distributions

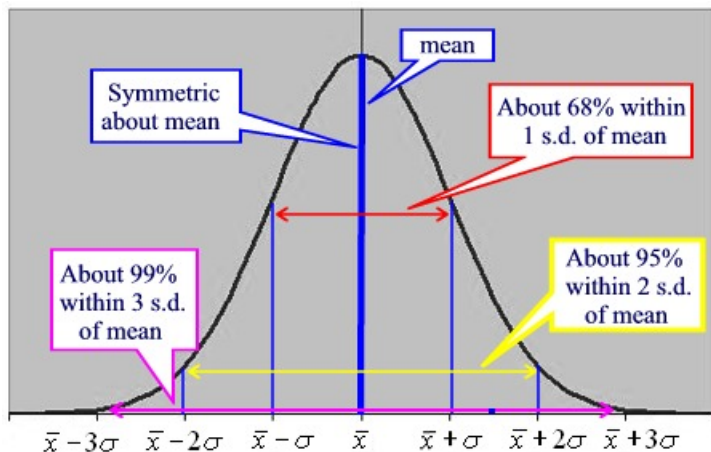


Figure C: Normal Distribution

Some Important Distributions

The Standard Normal Distribution

- Random variables that have a standard normal distribution are denoted by Z , and the standard normal cumulative distribution function is denoted by Φ . For example, $Pr(Z \leq c) = \Phi(c)$.
- The standard normal distribution is a very convenient representation for a random variable that is normally distributed because its values have been tabulated.
- The standard normal is important because if $X \sim N(\mu_X, \sigma_X^2)$ then $Z = \frac{X - \mu_X}{\sigma_X} \sim N(0, 1)$.
- Similarly, if $Z \sim N(0, 1)$ then $X = \mu_X + \sigma_X Z \sim N(\mu_X, \sigma_X^2)$

Some Important Distributions

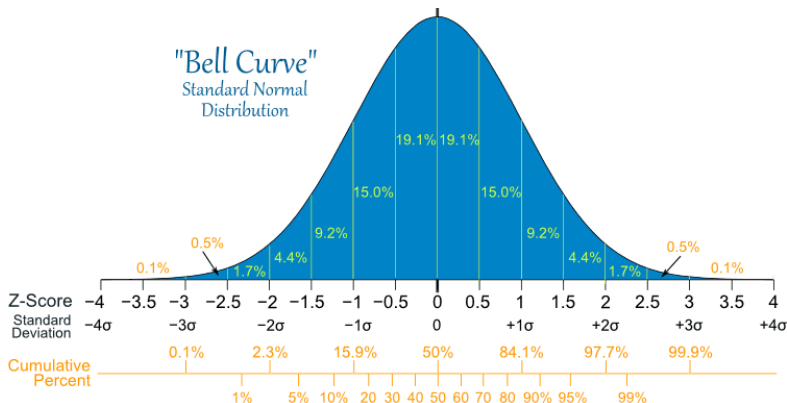


Figure C: Standard Normal Distribution

Some Important Distributions

The Multivariate Normal Distribution

The normal distribution can be generalized to represent the joint distribution of two (bivariate normal) or more than two (multivariate normal). The multivariate normal distribution has four important properties

Some Important Distributions

The Multivariate Normal Distribution: Property I

If X and Y have bivariate normal distribution with covariance σ_{XY} , then for constants a and b

$$aX \pm bY \sim N(a\mu_X \pm b\mu_Y, a^2\sigma_X^2 + b^2\sigma_Y^2 \pm 2ab\sigma_{XY}). \quad (33)$$

The Multivariate Normal Distribution: Property II

If a set of variables has a multivariate normal distribution, then the marginal distribution of each of the variables is normal.

Some Important Distributions

The Multivariate Normal Distribution: Property III

If variables with multivariate normal distribution have covariances that equal zero, then the random variables are independent. For example, if X and Y have a bivariate normal distribution with $\sigma_{XY} = 0$, then X and Y are independent. Here uncorrelation implies independence (not true in general).

The Multivariate Normal Distribution: Property IV

If X and Y have a bivariate normal distribution, then the conditional expectation of Y given X is linear in X that is $E[Y|X = x] = a + bx$, where a and b are constant. Joint normality implies linearity of conditional expectations, but linearity of conditional expectation does not imply joint normality.

Some Important Distributions

The Chi-Squared Distribution

It is the sum of M square independent standard normal random variables. This distribution depends on M , which is called the degrees of freedom of the chi-squared distribution, and it is denoted by χ_M^2 . For example, let Z_1 , Z_2 and Z_3 , be independent standard normal random variables. Then, $Z_1^2 + Z_2^2 + Z_3^2$ has a Chi-squared distribution with 3 degrees of freedom.

Some Important Distributions

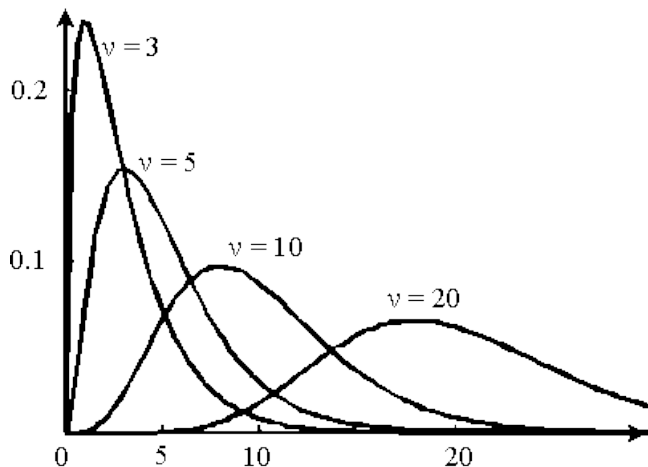


Figure C: Chi squared distribution

Some Important Distributions

The Student t Distribution

It is the ratio of a standard normal random variable, divided by the square root of an independently distributed chi-squared random variable with M degrees of freedom divided by M . This is

$$\frac{Z}{\sqrt{\frac{W}{M}}} \sim t_M. \quad (34)$$

Some Important Distributions

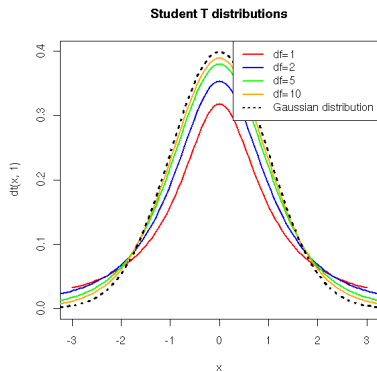


Figure C: Student t Distribution

Some Important Distributions

The F Distribution

The F distribution with M and N degrees of freedom, denoted by $F_{M,N}$ is defined to be the distribution of the ratio of a chi-squared random variable with M degrees of freedom, divided by M , to an independent chi-squared distribution with N degrees of freedom, divided by N .

Some Important Distributions

The F Distribution

Let W be a χ_M^2 and V be χ_N^2 and suppose W and V are independently distributed, then

$$\frac{\frac{W}{M}}{\frac{V}{N}} \sim F_{M,N}. \quad (35)$$

When the degrees of freedom in the denominator N is large, the distribution of the random variable converges to that of a χ_M^2

Some Important Distributions

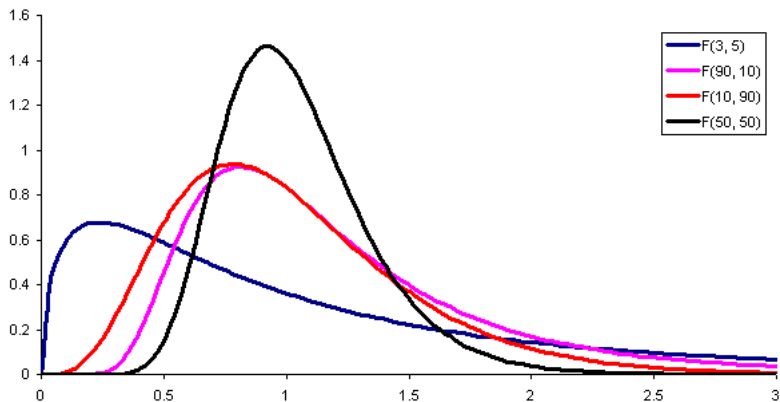


Figure C: F Distribution

Random Sampling and the Distribution of the Sample Average

Random Sampling

- Most of the procedures used in econometrics make use of averages of sample data, and thus it is important to characterize the distribution of these sample averages.
- In a simple random sampling procedure n objects are selected at random from a population, and each member of the population is equally likely to be included in the sample.
- We will denote the n observations in the sample by $Y_1, \dots, Y_n$, where Y_1 is the first observation, Y_2 is the second observation, and so forth. Each of these Y_i 's are random variables.

Random Sampling and the Distribution of the Sample Average

Random Sampling

- Because $Y_1, \dots, Y_n$ are randomly drawn from the same population, the marginal distribution of Y_i is the same for each $i = 1, \dots, n$; this marginal distribution is the distribution of Y in the population being sampled.
- When each Y_i has the same marginal distribution for all i , then the Y_i 's are said to be identically distributed.

Random Sampling and the Distribution of the Sample Average

Random Sampling

- Under simple random sampling, knowing the value of Y_1 provides no information about Y_2 , so the conditional distribution of Y_2 given Y_1 is simply the marginal distribution of Y_2 .
- In other words, under simple random sampling the Y_i 's are independent of one another.

Random Sampling and the Distribution of the Sample Average

Random Sampling

- When $Y_1, \dots, Y_n$ are drawn from the same distribution and are independently distributed, they are said to be independently and identically distributed (or, i.i.d.)

Random Sampling and the Distribution of the Sample Average

The Sample Average or Sample Mean

The sample average or sample mean, $\bar{Y}$, of the n observations $Y_1, \dots, Y_n$ is

$$\bar{Y} = \frac{1}{n}(Y_1 + \dots + Y_n) = \frac{1}{n} \sum_{i=1}^n Y_i. \quad (36)$$

Note that the sample mean is a random variable, and as such it has a probability distribution that we call the sampling distribution of $\bar{Y}$.

Random Sampling and the Distribution of the Sample Average

Mean and Variance of the Sample Average

Suppose that the observations $Y_1, \dots, Y_n$ are i.i.d., and let μ_Y and σ_Y^2 denote the mean and variance of each Y_i . The mean of the sample average is

$$E[\bar{Y}] = \mu_{\bar{Y}} = \mu_Y. \quad \text{Show it!} \quad (37)$$

The variance of the sample mean is

$$\text{Var}[\bar{Y}] = \sigma_{\bar{Y}} = \frac{\sigma_Y^2}{n}. \quad \text{Show it!} \quad (38)$$

Finally, the standard deviation of the sample mean is

$$\sigma_{\bar{Y}} = \frac{\sigma_Y}{\sqrt{n}}. \quad (39)$$

Random Sampling and the Distribution of the Sample Average

Distribution of the Sample Average

In sum, the mean, variance, and standard deviation of the sample mean are

$$E[\bar{Y}] = \mu_Y, \quad \text{Var}[\bar{Y}] = \frac{\sigma_Y^2}{n}, \quad \sigma_{\bar{Y}} = \frac{\sigma_Y}{\sqrt{n}}.$$

Random Sampling and the Distribution of the Sample Average

Normality of the Sample Average $\bar{Y}$

- When $Y_1, \dots, Y_n$ are i.i.d. draws from a population that is distributed $N(\mu_Y, \sigma_Y^2)$ the sample mean $\bar{Y}$ is distributed $N(\mu_Y, \frac{\sigma_Y^2}{n})$.
- When the distribution of Y is not normal, the exact distribution (finite sample distribution approach) of the sample mean is typically complicated and depends on the distribution of Y .
- An alternative approach looks at the asymptotic distribution (large sample approach).

Random Sampling and the Distribution of the Sample Average

Large Sample Approximations to Sampling Distributions

- The large sample approach uses approximations to the sampling distribution that rely on the sample being large enough ($n > 30$).
- The approximation based on this approach is also called asymptotic distribution, where the term "asymptotic" means that the approximation becomes exact in the limit as $n \rightarrow \infty$.
- There are two key results: *The Law of Large Numbers*, and *The Central Limit Theorem*.

Random Sampling and the Distribution of the Sample Average

The Law of Large Numbers

The Law of Large Numbers says that when the sample size is large, $\bar{Y}$ will be very close to μ_Y with very high probability.

The Central Limit Theorem

The Central Limit Theorem says that, when the sample size is large, the sampling distribution of the standardized sample average is approximately normal.

Random Sampling and the Distribution of the Sample Average

Asymptotic Theory

- While exact sampling distributions are complicated and depend on the distribution of Y , asymptotic distributions are simple.
- Most of the econometrics we will discuss in this class relies on this normal approximation.

Random Sampling and the Distribution of the Sample Average

Convergence in Probability

A sequence of random variables X_n converges in probability to X if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \Pr(|X_n - X| > \epsilon) = 0. \quad (40)$$

Law of Large Numbers: Formal Statement

Suppose that Y_i for $i = 1, \dots, n$ are i.i.d. with $E[Y_i] = \mu_Y$ and $\text{Var}[Y] = \sigma_Y^2 < \infty$ (large outliers are unlikely). Then, the sample average $\bar{Y}$ converges in probability to μ_Y (or equivalently $\bar{Y}$ is consistent for μ_Y) if the probability that the sample average $\bar{Y}$ is in the range $\mu_Y - c$ to $\mu_Y + c$ becomes arbitrarily close to 1 as n increases for any constant $c > 0$.

Random Sampling and the Distribution of the Sample Average

Central Limit Theorem

- Recall that the distribution of $\bar{Y}$ is exactly normal $N\left(\mu_Y, \frac{\sigma_Y^2}{n}\right)$ when the sample is drawn from a population with a normal distribution.
- The CLT says that for n large enough the distribution of $\bar{Y}$ is well approximated by the normal distribution above.
- The CLT is stated in terms of the standard normal distribution.

Central Limit Theorem: Formal Statement

Suppose that Y_i for $i = 1, \dots, n$ are i.i.d. with $E[Y_i] = \mu_Y$ and $\text{Var}[Y] = \sigma_Y^2 < \infty$. Thus, as $n \rightarrow \infty$ we have

$$\frac{\bar{Y} - \mu_Y}{\sigma_{\bar{Y}}} = \sqrt{n} \left[\frac{\bar{Y} - \mu_Y}{\sigma_Y} \right] \approx N(0, 1). \quad (41)$$

Some Important Distributions

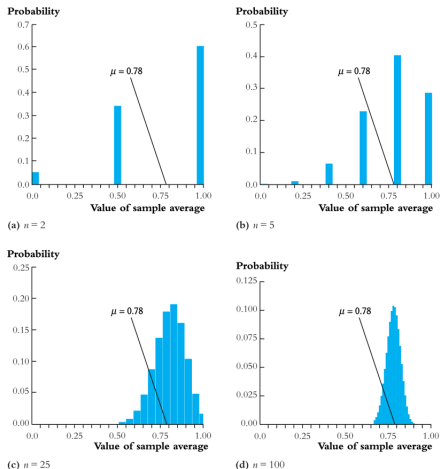


Figure C: Sampling Distribution of Sample Mean -Bernoulli $p = 0.78$ Case.

Some Important Distributions

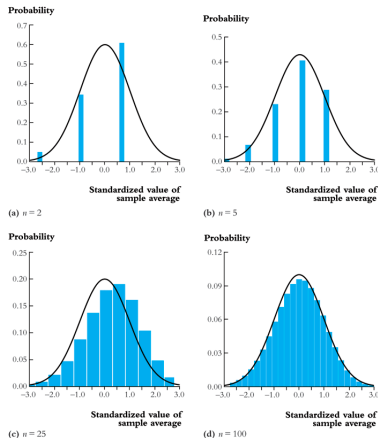


Figure C: Sampling Distribution of Standardized Sample Mean -Bernoulli $p = 0.78$ Case.

Review of Statistics

Statistics and Econometrics

- Statistics is the science of using data to learn about the world around us. We use statistics to answer questions about unknown characteristics of distributions in the population of interest.
- Comprehensive surveys are typically impractical. We need to learn about the population using a sample (ideally, a random one). Statistics will allow us to use this sample to draw conclusions about the population.

Review of Statistics

Statistics and Econometrics

In this class we will use three types of statistical methods:

- 1 Estimation.
- 2 Hypothesis testing.
- 3 Confidence intervals.

Estimation of the Population Mean

Estimators and Estimates

- An estimator is a function of a sample of data to be drawn from a population.
- An estimate is the numerical value of the estimator when it is actually computed using data from a specific sample.
- An estimator is a random variable because it is a function of the random sample observations while an estimate is a non-random number.

Estimation of the Population Mean

Estimators and their Properties

- A natural estimator for the population mean μ_Y is the sample mean $\bar{Y}$ from a random sample.
- However, there are multiple possible estimators of μ_Y , for example, Y_1 .
- What are the desirable characteristics of the sampling distribution of an estimator?

Estimation of the Population Mean

Property I: Unbiasedness

Let $\hat{\mu}_Y$ be an estimator of μ_Y . We say $\hat{\mu}_Y$ is an unbiased estimator of μ_Y if $E[\hat{\mu}_Y] = \mu_Y$. The bias of the estimator is $E[\hat{\mu}_Y] - \mu_Y$. In other words, if we compute the value of the estimator for different samples, on average we get the right number.

Estimation of the Population Mean

Property II: Consistency

Let $\hat{\mu}_Y$ be an estimator of μ_Y . We say $\hat{\mu}_Y$ is a consistent estimator of μ_Y if $\hat{\mu}_Y$ converges in probability to μ_Y . In other words, when the sample size is large, the uncertainty about the value of μ_Y arising from random variation in the sample is very small.

Convergence in Probability

A sequence of random variables $\{X_n\}$ converges in probability to X if for all $\epsilon > 0$

$$\lim_{n \rightarrow \infty} Pr(|X_n - X| > \epsilon) = 0. \quad (42)$$

Estimation of the Population Mean

Property III: Efficiency

Let $\hat{\mu}_Y$ and $\tilde{\mu}_Y$ be unbiased estimators of μ_Y . We say that $\hat{\mu}_Y$ is more efficient than $\tilde{\mu}_Y$ if $V[\hat{\mu}_Y] < V[\tilde{\mu}_Y]$. In other words, an estimator is more efficient than other if it has a tighter sampling distribution.

Estimation of the Population Mean

Properties of $\bar{Y}$

- The sample mean $\bar{Y}$ is an unbiased and consistent estimator of μ_Y .
- The sample mean $\bar{Y}$ is the best linear unbiased estimator (BLUE), where “best” stands for more efficient here.
- The sample mean $\bar{Y}$ is also the least squares estimator of μ_Y .

Hypothesis Tests Concerning the Population Mean

Hypothesis Tests About μ_Y

The objective is to make a provisional decision based on the evidence at hand whether a *null* hypothesis is true, or instead that some *alternative* hypothesis is true. That is, test

- ❶ $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] > \mu_{Y,0}$ (1-sided $>$).
- ❷ $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] < \mu_{Y,0}$ (1-sided $<$).
- ❸ $H_0 : E[Y] = \mu_{Y,0}$ versus $H_I : E[Y] \neq \mu_{Y,0}$ (2-sided).

Here, $\mu_{Y,0}$ denotes the hypothesised value of the population mean (i.e., the value of the population mean “under” the null hypothesis).

Hypothesis Tests Concerning the Population Mean

The p -value

The p -value is the probability of drawing a statistic at least as adverse to the null hypothesis as the value actually computed with your data, assuming that the null hypothesis is true. One often “rejects the null hypothesis” when the p -value is less than the significance level α .

The significance level α

The significance level of a test is a pre-specified probability of incorrectly rejecting the null, when the null is true.

Hypothesis Tests Concerning the Population Mean

Example

- Suppose we are analyzing the height of the population of students at Stony Brook University, and that we believe the average height is 1.75 mts. This is our null hypothesis.
- We draw a random sample and we find that the average height in the sample is 1.8 mts.
- The p-value is the probability of observing a sample mean $\bar{Y} = 1.8$ mts. when the true population average is 1.75 mts.
- If the p-value is small (for example 0.05), then it is very unlikely that a random sample would lead to an estimate of 1.8 mts when the true population value is 1.75 mts.

Hypothesis Tests Concerning the Population Mean

Calculation of the *p*-value

Let $\bar{Y}^{act}$ be the value of the sample average “actually” computed with the sample data and let Pr_{H_0} denote the probability computed under the null hypothesis (i.e., computed assuming that $E[Y] = \mu_{Y,0}$). Then, the p-value is

$$\text{p-value} = Pr_{H_0} \left(|\bar{Y} - \mu_{Y,0}| > |\bar{Y}^{act} - \mu_{Y,0}| \right) \quad \text{two-sided} \quad (43)$$

$$\text{p-value} = Pr_{H_0} \left((\bar{Y} - \mu_{Y,0}) > (\bar{Y}^{act} - \mu_{Y,0}) \right) \quad \text{one-sided} \quad (44)$$

Hypothesis Tests Concerning The Population Mean

Computing the p-value for n Large and σ_Y Known

$$\begin{aligned}\text{p-value} &= Pr_{H_0} \left(|\bar{Y} - \mu_{Y,0}| > \left| \bar{Y}^{act} - \mu_{Y,0} \right| \right), \\ &= Pr_{H_0} \left(\left| \frac{\bar{Y} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= Pr_{H_0} \left(Z_0 > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= 2\phi \left(- \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right).\end{aligned}\tag{45}$$

where $\sigma_{\bar{Y}} = \sigma_Y / \sqrt{n}$.

Hypothesis Tests Concerning The Population Mean

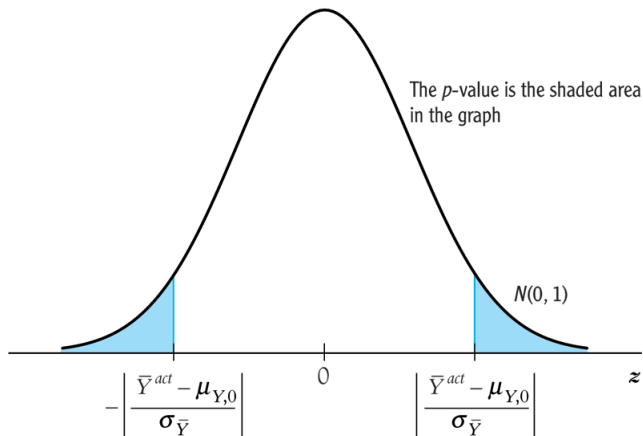


Figure C: p -value for known variance.

Hypothesis Tests Concerning the Population Mean

Example

- Suppose we know that the standard deviation in the distribution of height among Stony Brook University students is 0.3 mts (i.e., $\sigma_Y = 0.3$).
- In a sample of 100 students the computed sample average height is $\bar{Y} = 1.8$ mts.
- Compute the p-value for the tests
 - ① $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] > 1.75$ (1-sided $>$).
 - ② $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] \neq 1.75$ (2-sided).

Hypothesis Tests Concerning The Population Mean

Remarks

- However, to proceed in the way we suggested above, we still need to know the population variance.
- In general the population variance is unknown and it must be estimated before the p-value can be computed.

Hypothesis Tests Concerning the Population Mean

Sample Variance and Sample Standard Deviation

The sample variance s_Y^2 is

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2. \quad (46)$$

The sample standard deviation s_Y is the square root of the sample variance. The sample variance is an unbiased and consistent estimator of the population variance.

Hypothesis Tests Concerning the Population Mean

Standard Error of $\bar{Y}$

The standard error of $\bar{Y}$ is an estimator of the standard deviation of $\bar{Y}$ and is denoted by $SE(\bar{Y})$ or $\bar{s}$. When $Y_1, \dots, Y_n$ are i.i.d. we have

$$SE(\bar{Y}) = \bar{s} = \frac{s_Y}{\sqrt{n}}. \quad (47)$$

Hypothesis Tests Concerning The Population Mean

Computing the p-value for n Large and σ_Y Unknown

Since the sample variance is a consistent estimator of the population variance, the p-value can be computed by using $SE(\bar{Y})$. Thus, when σ_Y is unknown we can use the formula

$$\text{p-value} = 2\phi \left(- \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{SE(\bar{Y})} \right| \right), \quad (48)$$

where $SE(\bar{Y}) = \bar{s} = \frac{s_Y}{\sqrt{n}}$.

Hypothesis Tests Concerning The Population Mean

The t-statistic

The standardized sample average $(\bar{Y} - \mu_{Y,0}) / SE(\bar{Y})$ plays an important role in testing statistical hypotheses and it is called the *t-statistic* or *t-ratio*,

$$t = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}. \quad (49)$$

When n is large, the sample variance is close to the population variance, and thus the distribution of the t-statistic can be approximated by the standard normal.

Hypothesis Tests Concerning The Population Mean

The t-statistic and the p-value

We can write the formula for the p-value in terms of the t-statistic. Let t^{act} denote the value of the t-statistic actually computed

$$t^{act} = \frac{\overline{Y}^{act} - \mu_{Y,0}}{SE(\overline{Y})}. \quad (50)$$

Hence, when n is large the p-value can be computed by

$$\text{p-value} = 2\phi(-|t^{act}|) \quad \text{two-sided}, \quad (51)$$

$$\text{p-value} = 1 - \phi(t^{act}) \quad \text{one-sided } >, \quad (52)$$

$$\text{p-value} = \phi(t^{act}) \quad \text{one-sided } <. \quad (53)$$

Hypothesis Tests Concerning the Population Mean

Example

- Suppose the standard deviation in the distribution of height among Stony Brook University students is unknown (σ_Y unknown).
- In a random sample of 100 students the computed sample average height is $\bar{Y}^{act} = 1.8$ mts and the sample variance is $s^2 = 0.08$.
- Compute the p-value for the test $H_0 : E[Y] = \mu_{Y,0} = 1.75$ versus $H_I : E[Y] > 1.75$.

Hypothesis Tests Concerning the Population Mean

Hypothesis Testing With a Prespecified Significance Level

Type I Error

Reject the null hypothesis when it is true.

Type II Error

Not rejecting the null hypothesis when it is false.

Significance Level

The significance level of the test is the probability of rejecting the null hypothesis when it is true (probability of type I error). Hence, another way to think about the p-value is as the smallest significance level at which you can reject the null hypothesis.

Hypothesis Tests Concerning the Population Mean

Hypothesis Testing With a Prespecified Significance Level

Critical Value

The critical value of the test is the value of the test statistic for which the test just rejects the null hypothesis at the chosen significance level.

Rejection Region

The rejection region is the set of values of the test statistic for which the null hypothesis is rejected.

Acceptance Region

The acceptance region is the set of values of the test statistic for which the null hypothesis is not rejected.

Hypothesis Tests Concerning the Population Mean

Hypothesis Testing With a Prespecified Significance Level

Size of the Test

The size of the test is the probability of type I error.

Power of the Test

The power of the test is the probability of rejecting the null hypothesis when the alternative is true.

Hypothesis Tests Concerning the Population Mean

Hypothesis Testing With a Prespecified Significance Level

How Do We Choose the Significance Level?

- Econometricians typically use a 5% significance level, which means that they are incorrectly rejecting the null hypothesis on average 1 out of 20 times.
- Using a very low significance level makes more difficult to reject the null hypothesis when it is false (i.e., the lower the significance level the lower the power of the test).

Hypothesis Tests Concerning the Population Mean

Implementing a Two-Sided Test on the Population Mean

- 1 Compute $SE(\bar{Y})$ (standard error of $\bar{Y}$).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0}(Z > |t^{act}|) = 2\phi(-|t^{act}|).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if $|t^{act}| > 1.96$, where 1.96 is the critical value of the t statistic (i.e., the value that leaves 2.5% of the probability mass to the right).

Hypothesis Tests Concerning the Population Mean

Implementing a One-Sided ($>$) Test on the Population Mean

- 1 Compute $SE(\bar{Y})$ (standard error of $\bar{Y}$).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0} (Z > t^{act}) = 1 - \Phi(t^{act}).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if $t^{act} > 1.64$, where 1.64 is the critical value of the t statistic (i.e., the value that leaves 5% of the probability mass to the right).

Confidence Intervals for the Population Mean

- Even though we cannot learn the true value of μ_Y (the population mean of Y) using only a sample, we can construct an interval containing μ_Y with a certain prespecified probability.
- This interval is called *confidence interval*, and the probability level is called the *confidence level*.
- For example, a 95% confidence interval for μ_Y is an interval that contains the true value of μ_Y in 95% of repeated samples.

Confidence Intervals for the Population Mean

Some Confidence Intervals

When sample sizes are large the 90%, 95%, and the 99% confidence intervals are

$$90\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.64 \times SE(\bar{Y})] . \quad (54)$$

$$95\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.96 \times SE(\bar{Y})] . \quad (55)$$

$$99\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 2.58 \times SE(\bar{Y})] . \quad (56)$$

Hypothesis Tests Concerning the Population Mean

Remarks: What is Random Here?

- The values of $Y_1, \dots, Y_n$ and thus any function of them including the confidence interval which will differ from one sample to the next.
- The population parameter μ_Y is not random; we just don't know it.

Using the t-Statistic When the Sample Size is Small

The t-Statistic Testing the Mean

Let Y_i for $i = 1, \dots, n$ be an i.i.d. sample. As we have stated before, the t-statistic for testing $H_0 : E(Y) = \mu_{Y,0}$ is

$$t = \frac{\bar{Y} - \mu_{Y,0}}{\sqrt{s_Y^2/n}} = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}, \quad (57)$$

where $s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ is the sample variance.

Using the t-Statistic When the Sample Size is Small

The t-Statistic Testing the Mean

- When the sample size is small, the standard normal distribution is a poor approximation for the distribution of the t-statistic.
- The exact distribution of the t-statistic is complicated and depends on the distribution of Y .
- However, if the population distribution is normal, the exact distribution of the t-statistic testing the mean of a single population is the *Student t* distribution with $n - 1$ degrees of freedom.

Using the t-Statistic When the Sample Size is Small

Some Remarks on the Use of the Student t Distribution in Practice

- For testing the mean of Y , the *Student t Distribution* can be used **only** if the underlying population distribution is normal.
- If the sample size is moderate (several dozen) or large (hundreds or more) the difference between the t -Distribution and the Standard Normal critical values is negligible.

Hypothesis Tests Concerning The Population Mean

degrees of freedom ($n - 1$)	5% t -distribution critical value
10	2.23
20	2.09
30	2.04
60	2.00
∞	1.96

Figure C: Critical values in the t -Student Distribution

Using the t-Statistic When the Sample Size is Small

Some Remarks on the Use of the Student t Distribution in Practice

- In sum, the Student-t distribution is only relevant when the sample size is very small; but in that case, for it to be correct, you must be sure that the population distribution of Y is normal.
- In economic data, the normality assumption is rarely credible.

Using the t-Statistic When the Sample Size is Small

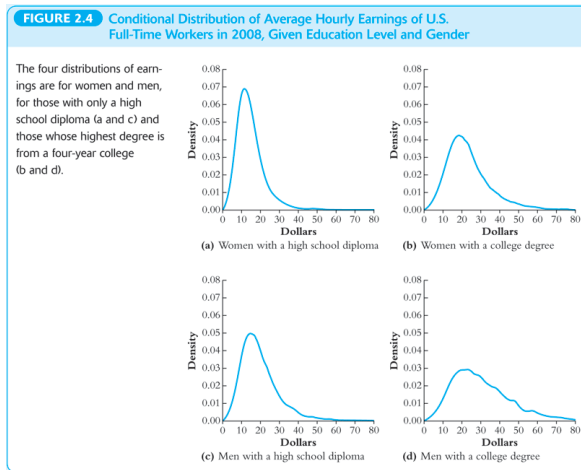


Figure C: Economic Data and the Normality Assumption

Scatterplots, the Sample Covariance, and the Sample Correlation

Scatterplots

A *scatterplot* is a plot of n observations on X_i and Y_i , in which each observation is represented by the point (X_i, Y_i) .

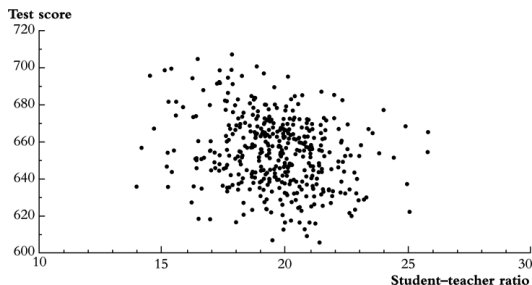


Figure C: Scatterplot: Test Scores and Student-Teacher Ratio

Scatterplots, the Sample Covariance, and the Sample Correlation

Sample Covariance and Sample Correlation

The sample covariance and correlation are estimators of their population counterparts.

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}) (Y_i - \bar{Y}) \quad \text{Sample Covariance} \quad (58)$$

$$r_{XY} = \frac{s_{XY}}{s_X s_Y} \quad \text{Sample Correlation} \quad (59)$$

where s_X and s_Y are the sample standard deviations of X and Y respectively.

Scatterplots, the Sample Covariance, and the Sample Correlation

Sample Covariance and Sample Correlation

- The sample covariance is a consistent estimator of the population covariance.
- The sample correlation coefficient measures the strength of the linear association between X and Y in the sample of n observations.
- The sample correlation lies between -1 and 1 .

Scatterplots, the Sample Covariance, and the Sample Correlation

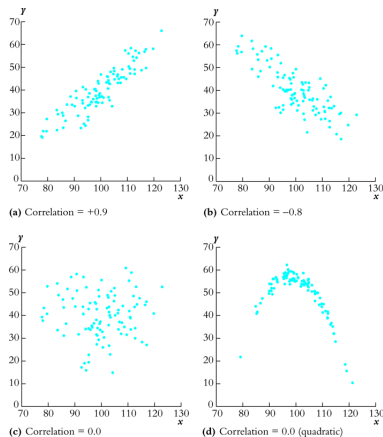


Figure C: Scatterplots and Correlation